

Electronic Eye SECURITY SYSTEM

PAMARTHI KANAKRAJA

The electronic eye security systems available in the market are expensive, and programming is very difficult for a micro-controller based system. Here is a simple and inexpensive electronic eye security system that you can build easily.

Main objective of this project is to design a security system based on a photo-sensing arrangement. It uses dual comparator LM393 that is widely used in many commercial applications. Here, it is used to sense the input signal through a light-dependent resistor (LDR). Comparator output drives a buzzer

and a relay.

This concept is very useful as a burglar deterrent in places like banks, malls, jewellery stores, hotels and homes. Fig. 1 shows the author's prototype.

Circuit and working

Circuit diagram for the electronic eye security system is shown in Fig. 2. It is built around 12V-0-12V, 750mA secondary transformer (X1), 1N4007 rectifier diodes (D1 through D3), LDR (LDR1), filter capacitors C1 and C2, 5V voltage regulator 7805 (IC1), dual comparator LM393 (IC2), 12V multi-tone high-gain buzzer (PZ1), transistor TIP122 (T1) for driving PZ1, transistor BC547 (T2) for driving 12V, 1C/O relay for controlling an electrical appliance or bulb (B1), and 10-kilo-ohm potmeter VR1.

When light falls on LDR1, its resistance drastically decreases, which leads to triggering an alarm to alert the user.

This system is suitable for securing cash boxes and lockers. The circuit is placed inside the cash box or locker in such a way that, when the burglar opens it and uses a torch light to find the valuables, light falls on LDR1, which acts as an electronic eye.

PARTS LIST

Semiconductors:

- IC1 - 7805 5V regulator
- IC2 - LM393 dual comparator
- T1 - TIP122 power transistor
- T2 - BC547 npn transistor
- D1-D3 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R4 - 1-kilo-ohm
- VR1 - 10-kilo-ohm potmeter

Capacitors:

- C1 - 1000 μ F, 40V electrolytic
- C2 - 0.1 μ F ceramic disk

Miscellaneous:

- LDR1 - Light dependent resistor
- RL1 - 12V, 1C/O relay
- B1 - 40W, 230V AC bulb
- PZ1 - 12V multi-tone high-gain buzzer
- CON1 - 2-pin connector
- X1 - 230V AC primary to 12V-0-12V, 750mA secondary transformer
- 3-pin connector for X1
- 8-pin IC base
- 2-pin power cord for AC mains

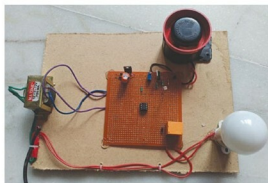


Fig. 1: Author's prototype

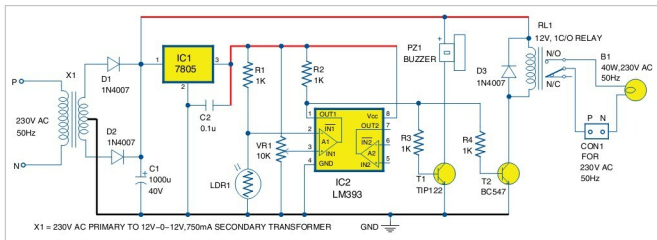


Fig. 2: Circuit diagram of the electronic eye security system